Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Practical – III TASK

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Task (on house_price_regression_dataset.csv) (Don't include in print document)

Q.1 Find the correlation between Year_Built and House_Price.

Input:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
```

```
print("First 5 rows of dataset:")
print(df.head())
correlation = df["Year_Built"].corr(df["House_Price"])
print("\nQ1. Correlation between Year_Built and House_Price:")
print(correlation)
```

Output:-

```
First 5 rows of dataset:
   Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  Lot_Size  \
0            1360             2              1        1981  0.599637
1            4272             3              3        2016  4.753014
2            3592             1              2        2016  3.634823
3             966             1              2        1977  2.730667
4            4926             2              1        1993  4.699073

   Garage_Size  Neighborhood_Quality   House_Price
0            0                     5  2.623829e+05
1            1                     6  9.852609e+05
2            0                     9  7.779774e+05
3            1                     8  2.296989e+05
4            0                     8  1.041741e+06
```

```
Correlation between Year_Built and House_Price:
5196736193106453
```

Q.2 Create a correlation matrix for all numerical columns.

INPUT,OUTPUT:-

```
print("\nQ2. Correlation Matrix:")
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)
```

```
Q2. Correlation Matrix:
                     Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  \
Square_Footage             1.000000     -0.043564      -0.031584   -0.022392
Num_Bedrooms              -0.043564      1.000000       0.022848   -0.015820
Num_Bathrooms             -0.031584      0.022848       1.000000   -0.021063
Year_Built                -0.022392     -0.015820      -0.021063    1.000000
Lot_Size                   0.089479     -0.009355       0.034923   -0.061050
Garage_Size                0.030593      0.113761       0.024846   -0.025485
Neighborhood_Quality      -0.008357     -0.049024       0.017585   -0.009549
House_Price                0.991261      0.014633      -0.001862    0.051967

                     Lot_Size  Garage_Size  Neighborhood_Quality  House_Price
Square_Footage       0.089479     0.030593             -0.008357     0.991261
Num_Bedrooms        -0.009355     0.113761             -0.049024     0.014633
Num_Bathrooms        0.034923     0.024846              0.017585    -0.001862
Year_Built          -0.061050    -0.025485             -0.009549     0.051967
Lot_Size             1.000000     0.002436              0.037630     0.160412
Garage_Size          0.002436     1.000000             -0.011287     0.052133
Neighborhood_Quality 0.037630    -0.011287              1.000000    -0.007770
```

```
def regression_model(x_column, y_column):
    X = df[[x_column]]
    Y = df[y_column]

    model = LinearRegression()
    model.fit(X, Y)

    prediction = model.predict(X)
    r2 = r2_score(Y, prediction)

    return model.coef_[0], model.intercept_, r2
```

Q.3 Perform linear regression using:
X = Year_Built
Y = House_Price
Display the coefficient and intercept.
INPUT,OUTPUT:-

```
coefficient, intercept, r2_year = regression_model(
    "Year_Built", "House_Price"
)

print("\nQ3. Linear Regression: Year_Built -> House_Price")
print("Coefficient:", coefficient)
print("Intercept:", intercept)
```

```
Q3. Linear Regression: Year_Built -> House_Price
Coefficient: 638.6524884727302
Intercept: -649854.0823295021
```

Q.4 Calculate the R^2 score for the regression model: Year_Built - House_Price

INPUT,OUTPUT:-

```
print("\nQ4. R2 Score: Year_Built -> House_Price")
print(r2_year)
```

```
Q4. R2 Score: Year_Built -> House_Price
0.0027006067060743044
```

Q.5 Calculate the R^2 score for:

Square_Footage - House_Price

INPUT,OUTPUT:-

```
coefficient_sq, intercept_sq, r2_square = regression_model(
    "Square_Footage", "House_Price"
)

print("\nQ5. R2 Score: Square_Footage -> House_Price")
print(r2_square)
```

```
Q5. R2 Score: Square_Footage -> House_Price
0.9825992634147099
```

Q.6 Calculate the R^2 score for:

Neighborhood_Quality - House_Price

INPUT,OUTPUT:-

```
coefficient_nq, intercept_nq, r2_quality = regression_model(
    "Neighborhood_Quality", "House_Price"
)

print("\nQ6. R2 Score: Neighborhood_Quality -> House_Price")
print(r2_quality)
```

```
Q6. R2 Score: Neighborhood_Quality -> House_Price
6.037338916142776e-05
```

Q.7 Compare the R^2 scores of all three regression models.
INPUT,OUTPUT:-

```
print("\nQ7. Comparison of R2 Scores:")

print("Year_Built          :", r2_year)
print("Square_Footage      :", r2_square)
print("Neighborhood_Quality:", r2_quality)
```

```
Q7. Comparison of R2 Scores:
Year_Built          : 0.0027006067060743044
Square_Footage      : 0.9825992634147099
Neighborhood_Quality: 6.037338916142776e-05
```

```
cores = {
    "Year_Built": r2_year,
    "Square_Footage": r2_square,
    "Neighborhood_Quality": r2_quality
}

best = max(cores, key=cores.get)

print("\nBest predictor of House_Price:", best)
print("Highest R2 Score:", cores[best])
```

```
Best predictor of House_Price: Square_Footage
Highest R2 Score: 0.9825992634147099
```